C S Vineeth

AI / Machine Learning Engineer

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EDUCATION

BMS Institute of Technology and Management Bengaluru, India

B.E., Artificial Intelligence and Machine Learning 2019 – 2023

EXPERIENCE

CBK Infotech India Pvt. Ltd. Bengaluru, India

Technical Consultant Apr 2023 – Present

Cybersecurity & IoT Anomaly Detection

- Built leakage-safe intrusion-detection pipelines (211K+ flows, 44 features; Edge-IIoTset) with automated feature selection and SHAP validation, serving Random Forest, XGBoost, and LightGBM models at AUROC > 0.999.
- Developed streaming anomaly-detection systems over 2.2M+ sensor readings using LSTM and Transformer autoencoders (221,976 windows) with adaptive thresholding and simulated real-time inference for zero-day, edge-style detection.

Fraud Detection & Risk Intelligence

- Engineered production-grade fraud-scoring pipelines over 500K+ transactional and relational records with LightGBM, XGBoost, TabTransformer, and graph models, automating feature construction, class-imbalance handling, and threshold calibration to reach ROC-AUC 0.9995 and F1 0.986.
- Built reproducible batch-inference and model-benchmarking workflows across classical ML, tabular transformers, and GNNs, with a blockchain-backed audit prototype (model fingerprinting, input hashing) for traceable, decision-ready scoring.

Medical Imaging & Computer Vision

- Built and validated YOLO-based dental-detection pipelines (Precision 0.9843, Recall 0.9820, mAP@50 0.9861) with embedding-based similarity matching, automating preprocessing, training, and evaluation.
- Productionized transfer-learning workflows across diabetic-retinopathy grading, osteoporosis classification, EEG seizure detection, and edge-deployed gait analysis, standardizing reusable training and inference pipelines.

Genomic & Biomedical ML

- Built a scalable ClinVar data pipeline processing 8.9M raw variants into a clean 637K-variant dataset, automating review-status filtering, conflict resolution, SNV selection, and allele-validity checks.
- Engineered reusable 54-feature genomic matrices (encodings, transition/transversion, GC/CpG, k-mers, functional scores) with imputation and scaling for downstream deep-learning and ML benchmarking.

TCR Innovation Mumbai, India

AI Research Intern Nov 2022 – Jan 2023

- Developed computer-vision models for object counting, defect detection, and segmentation, improving detection accuracy 17% over baseline.
- Deployed inference workflows on AWS EC2/S3 with batching and caching, reducing latency 35% and infrastructure cost 22%.
- Built reusable preprocessing, training, and evaluation scripts to accelerate prototype iteration.

SKILLS

Languages & Data: Python, SQL, Pandas, NumPy

ML Engineering & Deployment: Batch & real-time inference APIs, model serving, pipeline automation, AWS (EC2/S3/IAM), TorchScript, ONNX, Vercel

Frameworks: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM

Modeling: CNNs, YOLO, LSTM, Transformers, Autoencoders, GNNs

Practices: Model benchmarking, reproducible pipelines, inference optimization, leakage-safe validation